Chemistry · Unit 1A · Level 2 (modified)

Charges & Static



TODAY'S PLAN

- ⚡ Initial model of lightning
- 📊 Correlation or cause?
- ✨ Sparks
- ⚛️ Particle model of static
- 🎈 Cause of static

Review · 5 minutes

Quick Review



🔬 Scientists use **models** to explain what they see.

⚡ Unit question: What causes lightning, and why are some places safer than others?

❓ What do you already know about lightning?

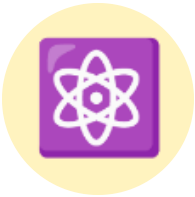
Words to Know

Words to Know



Lightning

A giant electric spark between a cloud and the ground (or two clouds).



Atom

The tiny particle that all matter is made of.



Proton

A positive (+) particle in the center (nucleus) of an atom. It does not move.



Electron

A negative (−) particle around the nucleus. Electrons can move from one object to another.

Words to Know

More Words to Know



⚪ **Neutron**: A particle in the nucleus with no charge (neutral).

🎈 **Static charge**: Extra or missing electrons that build up on an object, like a rubbed balloon.

⚖️ **Neutral**: An object with the same number of protons and electrons, so it has no overall charge.

🔗 **Correlation**: Two things happen together. It does not prove one causes the other.

Learn

Lightning: Our First Model

⚡ Lightning is a giant **electric spark** in a storm.

🌩️ It happens between **clouds** and the **ground**, or cloud to cloud.

🔥 It is hotter than the sun's surface and causes **thunder**.



Learn

Correlation vs. Cause

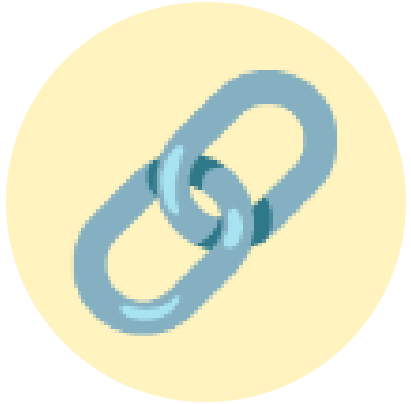


📊 Clouds with lots of **rain** also have lots of **lightning**.

🔗 That is a **correlation**: they happen together.

❓ It does not prove that rain **causes** lightning. We need more evidence.

🃏 Check 1 of 4



Two things happening together is called a...

Hold up 1, 2, or 3:

1️⃣ Cause

2️⃣ Correlation

3️⃣ Charge



Correlation

Correlation means they happen together. Cause needs more evidence.

Learn

Parts of an Atom



Proton

\+ charge, in the nucleus



Neutron

no charge, in the nucleus



Electron

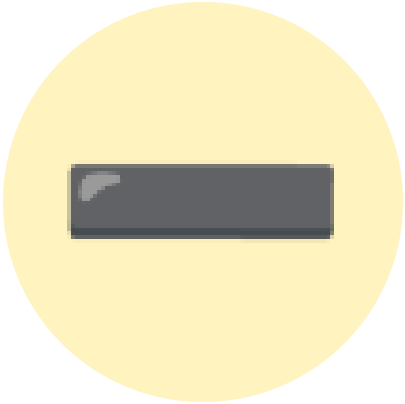
\- charge, moves around



Neutral atom

same number of + and -

🃏 Check 2 of 4



Which particle can move from one object to another?

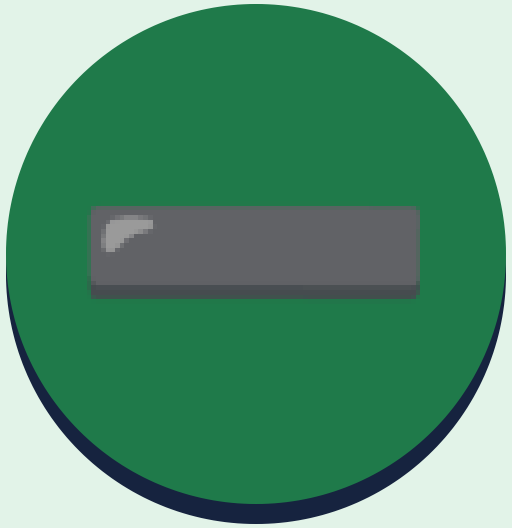
Hold up 1, 2, or 3:

1️⃣ Proton

2️⃣ Neutron

3️⃣ Electron

✅ Answer · Card 2 of 4



Electron

Electrons are on the outside and can move.

Learn

Particle Model of Static

🎈 Rubbing a balloon on hair moves **electrons** to the balloon.

➖ The balloon gets **extra electrons**: negative charge.

➕ The hair **loses electrons**: positive charge.

🧲 Opposite charges **attract**, so the hair sticks up to the balloon.



🃏 Check 3 of 4



After rubbing, the hair is positive because it...

Hold up 1, 2, or 3:

1️⃣ Lost electrons

2️⃣ Gained protons

3️⃣ Got wet

✅ Answer · Card 3 of 4



Lost electrons

Protons do not move. The hair lost electrons.

Learn

Sparks and Lightning

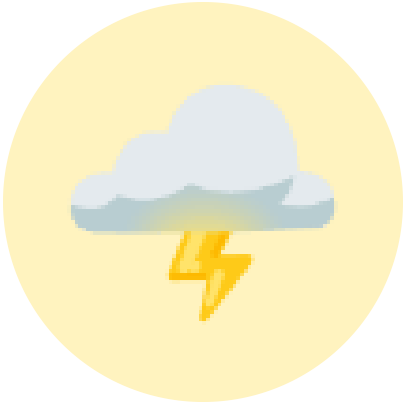


✨ When enough charge builds up, electrons **jump** through the air: a **spark**.

🌩️ In a storm cloud, moving ice and water rub and **separate charge**.

⚡ Lightning is the same idea, just **huge**.

🃏 Check 4 of 4



In a storm cloud, charge separates because…

Hold up 1, 2, or 3:

1️⃣ The sun heats it

2️⃣ Wind adds protons

3️⃣ Ice and water bump and rub

✅ Answer · Card 4 of 4

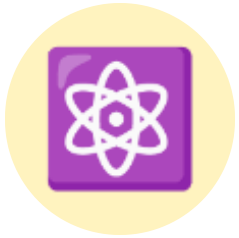


Ice and water bump and rub

Collisions move electrons, like rubbing a balloon.

Wrap-Up

Recap



Atom
protons, neutrons, electrons



Electrons
the only part that moves



Static
extra or missing electrons



Spark
charge jumps

Wrap-Up

Exit Ticket



1️⃣ Name the 3 parts of an atom and their charges.

2️⃣ Why does a rubbed balloon have a negative charge?

3️⃣ How is lightning like a spark?